

2.1 Integration Worksheet

1. Find $\int x e^{-3x} dx$.

2. Find $\int \frac{\ln(x)}{x^2} dx$

3. Find $\int x^2 \cos(x) dx$.

4. $\int \ln(x^2 + 1) dx$

5. $\int (x^2 + x + 1) \sin(x) dx$

6. Find $\int_{\sqrt{e}}^e \frac{\ln(x)}{x^2} dx$.

7. $\int x^3 \sqrt{2x+1} dx$

8. Calculate the exact value of: $\int_0^1 \tan^{-1}(x) dx$

9. Find $\int \cos(\ln(x)) dx$

10. $\int 4x^4 \sin(2x) dx$

11. Find $\int x \tan^2(x) dx$

12. Calculate the exact value of: $\int_{-1}^0 x \cdot 2^x \, dx$.

13. Find $\int e^x \sin(2x) \, dx$.